

# PocketSmart AI

## Project Documentation

### 1. Introduction

PocketSmart AI is a GenAI-powered budget and recommendation assistant designed to help users plan home decor, parties and jewelry purchases according to their budget and preferences.

### 2. Problem Statement

Users often spend considerable time searching for suitable products and planning events within a fixed budget. PocketSmart AI provides personalized recommendations to simplify this process.

### 3. Objectives

- Provide budget-based recommendations.
- Support home decor, party and jewelry planning.
- Reduce search and comparison time.
- Provide a simple and responsive web interface.
- Use Generative AI for personalized recommendations.

### 4. Key Features

- User registration and login
- Home Decor Planner
- Party Planner
- Jewelry Planner
- Budget-based recommendations
- AI-generated recommendations
- Fallback recommendations
- Recommendation results with estimated cost

### 5. System Architecture

The system consists of:

- **Frontend:** HTML, CSS, JavaScript and Jinja2
- **Backend:** Python and FastAPI

- **AI Service:** Google Gemini API
- **Database:** SQLite
- **Version Control:** GitHub

## 6. Application Modules

### Authentication Module

Handles user registration, login and logout.

### Home Planner

Provides recommendations based on budget, room type, style and requirements.

### Party Planner

Provides recommendations based on budget, event type, number of guests and theme.

### Jewelry Planner

Provides recommendations based on budget, occasion and jewelry type.

### Recommendation Service

Processes user requirements and generates suitable recommendations.

### Fallback Service

Provides predefined recommendations when the AI service is unavailable.

## 7. Application Workflow

User → Login/Register → Select Planner → Enter Requirements → Validate Input → Generate Recommendations → Display Results.

## 8. Database

SQLite is used for storing user information and application data. The main user fields include ID, name, email and password.

## 9. Testing

The application was tested for registration, login, planner functionality, budget input, recommendation generation, invalid input handling and fallback recommendations.

## 10. Limitations

- Recommendations depend on available AI services and catalogue data.
- Real-time product inventory and pricing are not guaranteed.
- Online purchasing and payment are not included.

## **11. Future Enhancements**

- Real-time marketplace integration
- Advanced recommendation personalization
- Recommendation history dashboard
- Improved authentication and security
- More planning categories

## **12. Conclusion**

PocketSmart AI provides a practical web-based solution for budget planning and personalized recommendations. The modular architecture allows the system to be maintained and extended with additional features in the future.